

Facet-Conditioned Trace Audits for Diagnosing Synthetic Persona Failures

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Abstract

Synthetic personas increasingly act from long-lived constructed identities rather than single-turn prompts. Their failures often include epistemic components: a system may omit a persona constraint, overweight a conflicting cue, express confidence unsupported by its construction, or provide a plausible but unfaithful justification. This paper introduces the Agentic Epistemology Framework (AEF), a trace schema and audit service for diagnosing such failures. AEF records selected facets, prompt provenance, facet weights, expected behavioral markers, response evidence, calibration checks, and optional counterfactual probes. We extend the framework with *facets*: persistent psychological, socio-ecological, psychographic, and interaction constraints that compose into an agent’s active frame. The reference implementation provides deterministic public harnesses that illustrate which trace fields are needed to localize injected failure classes, plus a preliminary live integration with a Vurvey persona-generation pipeline. In controlled synthetic harnesses, richer traces recover more injected labels than prompt-only or event-only traces; these numbers are best interpreted as trace-sensitivity checks rather than external validity estimates. A small live counterfactual study shows measurable output sensitivity to facet removal, but also reveals the main current limitation: expected-marker metadata is too sparse to diagnose real responses robustly. The contribution is therefore a reproducible audit interface, benchmark scaffold, and implementation path toward stronger validation of facet-conditioned persona faithfulness.

1 Introduction

Autonomous agents now plan, use tools, communicate, and revise internal state across long-running interactions. Synthetic personas add another layer: they must remain faithful to a constructed identity while responding naturally to novel stimuli. In both cases, important causal factors are often hidden inside prompt text, model activations, unstructured logs, or generated explanations that may not faithfully describe why the model produced a response.

This paper introduces the Agentic Epistemology Framework (AEF), a structured architecture for auditing agent reasoning. AEF represents beliefs as propositions with confidence and justifications, records observer events for epistemic transitions, and models frames as active cognitive perspectives that influence evidence weighting, source trust, and action thresholds. The paper further extends frames with *facets*: typed persona constraints that jointly shape the active frame. Facets can represent psychological dispositions, lived-context constraints, psychographic motivations, and interaction style. In a synthetic human simulation, facets are not independent prompt fragments. They interact nonlinearly, and those interactions should be inspectable.

This first publication of AEF has two goals. First, it defines a vocabulary and implementation pattern for epistemic auditability in agents and synthetic societies. Second, it provides executable benchmarks that test whether the resulting traces make known failure causes diagnosable. The framework does not claim that generated persona behavior is valid in the world merely because the system produces plausible text. Instead, it asks whether the system can expose enough structured evidence to debug when a persona fails to follow its own construction.

1.1 Contributions

This paper makes four contributions:

1. A structured epistemic ontology for beliefs, confidence, justifications, frames, facets, perceptions, actions, and observer events.
2. A facet composition model in which persistent persona constraints contribute nonlinearly to an agent’s active frame.
3. A TypeScript reference implementation with automated root-cause and facet-faithfulness benchmarks.
4. An evaluation protocol that combines structured traces with confidence calibration, black-box perturbation tests, and optional representation-level probes for reasoning faithfulness.

2 Background and Motivation

AEF draws from belief-desire-intention models [Bratman, 1987, Rao and Georgeff, 1991, Wooldridge, 2009], epistemic logic [Fagin et al., 2003], belief revision [Alchourrón et al., 1985, Gärdenfors, 1988], frame semantics [Minsky, 1974, Fillmore, 1976, Lakoff, 1987], argumentation theory [Dung, 1995, Nute, 1994], and agent-based modeling [Epstein and Axtell, 1996]. These traditions provide useful structure, but current LLM-based agents create additional audit problems.

Recent work shows that natural-language explanations should not be treated as automatically faithful. Chain-of-thought explanations can systematically misrepresent the true reasons for a model’s answer [Turpin et al., 2023], and faithfulness requires intervention-style tests rather than surface plausibility alone [Lanham et al., 2023]. Calibration and self-evaluation research shows that language models can sometimes estimate whether their own answers are correct, but this depends strongly on elicitation format and task setting [Kadavath et al., 2022]. Black-box hallucination detection methods, such as sampling-based consistency checks, provide another way to test whether outputs are stable under repeated generation [Manakul et al., 2023]. Meanwhile, representation engineering and sparse autoencoder work provide tools for probing latent directions or interpretable features when model activations are available [Zou et al., 2023, Cunningham et al., 2023].

AEF treats these lines of work as complementary. Observer traces expose the explicit state the system is designed to use. Black-box perturbations test whether outputs change when facets, evidence, or prompts are altered. Calibration tests ask whether the system knows when its response is likely faithful. Representation-level probes, when available, can test whether latent activations align with the asserted frame or facet. The framework is designed to hold all of these evidence types in a single audit trail.

3 Facet-Conditioned Synthetic Personas

Synthetic human simulation treats a persona as a complex adaptive system rather than a flat list of demographic labels. A persona has internal psychological dispositions, lived-context constraints, motivations, trust heuristics, resource constraints, and linguistic interaction style. In the Vurvey implementation inspected for this work, these variables are represented as a graph of PM facets. A facet is an axis of variation such as income, response style, source trust, or domain expertise. A facet value contains stable identifiers, prompts, taxonomy metadata, and reliability fields. Personas link to selected values through persona fragments. Population molds define facet distributions, mandatory and preferred values, weights, selection modes, and constraints.

This implementation suggests a natural extension to AEF. A **facet** is a typed, persistent constraint with fields:

$$Facet_i = (code_i, value_i, domain_i, prompt_i, weight_i, confidence_i, provenance_i). \quad (1)$$

The active frame is composed from the current stimulus x_t , the persona state S_t , and selected facets $Facet_{1:n}$:

$$F_t = g(S_t, x_t, Facet_{1:n}, I_{1:n}), \quad (2)$$

where $I_{1:n}$ denotes interaction terms between facets. This form is intentionally nonlinear. For example, high extraversion can produce very different behavior when combined with fragile self-confidence and extreme time poverty than when combined with high perceived control and abundant resources.

3.1 Facet Domains

The framework organizes facets into four audit domains:

1. **Psychological facets:** stable traits, anxiety level, self-confidence archetype, and emotional regulation constraints.
2. **Socio-ecological facets:** locale, life stage, socioeconomic status, time poverty, family structure, and other lived-context constraints.
3. **Psychographic facets:** motivations, adoption-curve position, source trust, perceived behavioral control, and decision heuristics.
4. **Interaction facets:** response style, conversational friction, objection tendency, and linguistic modulation.

Each facet should be auditable at three levels: whether it was selected, whether its prompt or structured representation reached the generation path, and whether the final response reflects its expected influence.

4 Epistemic State and Operational Rules

A **belief** is a proposition P held by an agent with confidence $conf \in [0, 1]$ and a justification. A **justification** is evidence, reasoning trace, source, or derivation path supporting a belief or response. A **frame** is the active cognitive lens that modulates interpretation and evidence weighting. A **facet** is a persistent constraint that contributes to frame composition. An **observer event** is a structured record of an epistemic transition.

AEF is guided by five operational rules:

1. **Traceability:** important actions and responses should trace back to beliefs, justifications, frames, and contributing facets.
2. **Explicit uncertainty:** beliefs and evaluator judgments should expose confidence rather than binary assertions.
3. **Frame awareness:** agents should record when frames change evidence weighting, source trust, or interpretation.
4. **Facet faithfulness:** persona responses should expose which facets were expected to constrain the response and which facets were actually expressed.
5. **Intervention readiness:** traces should support counterfactual tests such as removing a facet, swapping a facet value, or perturbing evidence source trust.

5 Confidence, Frame, and Facet Updates

Let $conf_{old}$ denote an agent’s current confidence in proposition P , and let e denote new evidence relevant to P . AEF uses update functions of the form:

$$conf_{new} = f(conf_{old}, just_{old}, e, F_t). \quad (3)$$

For frame-weighted evidence, evidence e suggests a target confidence $C(e, P) \in [0, 1]$. The active frame assigns evidence weight $w_{F_t}(e)$:

$$conf_{new} = (1 - w_{F_t}(e)) \cdot conf_{old} + w_{F_t}(e) \cdot C(e, P). \quad (4)$$

For facet-conditioned personas, $w_{F_t}(e)$ can be decomposed into facet contributions:

$$w_{F_t}(e) = \sigma \left(b + \sum_i \alpha_i h(Facet_i, e) + \sum_{i < j} \beta_{ij} h(Facet_i, Facet_j, e) \right), \quad (5)$$

where h maps facets and evidence to contribution scores, α_i are individual facet weights, β_{ij} are interaction weights, and σ bounds the resulting saliency. The observer should log individual contributions and interaction terms whenever the implementation can expose them.

6 Auditing Faithfulness

AEF distinguishes four kinds of faithfulness evidence.

Trace faithfulness asks whether the system recorded the inputs and transformations it was designed to use. For synthetic personas, this includes selected facets, facet values, prompt fragments, weights, constraints, and response markers.

Behavioral faithfulness asks whether the output reflects the selected persona constraints. This can be measured by entailment checks, contradiction checks, response-style classifiers, and facet-marker coverage.

Counterfactual faithfulness asks whether changing an input that should matter changes the output in the expected direction. Examples include swapping financial outlook from crisis flow to abundance, changing response style from declarer to verbatim repeater, or replacing peer-trust with institutional-trust.

Latent faithfulness asks whether hidden representations align with the claimed frame or facet when activations are available. Representation engineering can test directions corresponding to persona-relevant concepts, and sparse autoencoders can expose interpretable features. These probes are optional because many production systems expose only black-box access, but the framework leaves room for them as stronger evidence.

7 Reference Implementation

The reference implementation is written in TypeScript and is publicly available at <https://github.com/jroell/agent-epistemology-framework>. It implements beliefs, justifications, frames, agents, memory, actions, tools, and observer events. The implementation now includes two benchmark harnesses: a generic root-cause benchmark and a facet-frame faithfulness benchmark.

The Vurvey production codebase informed the facet extension. Its PM facet graph includes global facets and versioned facet values, `headerPrompt` and `detailPrompt` fields used to build persona prompts, `PmPersonaFragment` rows that link personas to selected facet values, mold facet configurations with weights and options, and auto-configuration services that recommend and weight facet distributions for population molds. The current implementation extends that path in two places: `vurvey-python` writes an AEF construction manifest into `ai_personas.facetConfigs` when personas are generated from molds, and `vurvey-api` attaches an AEF audit summary to each simulated survey answer during phase-two generation. These implementation details motivate the audit requirement: a persona response should be diagnosable back to the facet graph, prompt fragments, selected values, construction manifest, composed frame, rendered system prompt, user prompt, and generated response that produced it.

8 Automated Root-Cause Benchmark

The generic benchmark asks whether structured epistemic traces make it possible to identify the cause of agent failures in scenarios where the ground-truth cause is known. It covers four injected causes: frame misweighting, source-trust error, threshold miscalibration, and justification gap. It evaluates four trace conditions: full trace, no-frame trace, no-justification trace, and event-only trace.

We ran the benchmark with 160 synthetic cases using seed 42. The benchmark command was:

```
npx ts-node --transpile-only examples/root-cause-benchmark.ts --seed=42 --cases=160
```

9 Facet-Frame Faithfulness Benchmark

The second benchmark targets synthetic persona auditing. Each case constructs a persona from six facets: anxiety level, time poverty, financial outlook, source trust, response style, and adoption-curve position. The harness injects one of five failure types:

1. **Facet omission:** a required facet is absent from the composed frame.
2. **Facet misweighting:** a facet contribution is materially lower or higher than its expected calibration.

Table 1: Generic root-cause diagnosis results. Full traces expose the fields needed to recover all labels in this deterministic synthetic generator. Ablating frame fields, justification/source fields, or epistemic details removes diagnostic capability.

Trace condition	Accuracy	Macro F1	Coverage	Evidence inspected	Trace events
full	1.000	1.000	1.000	8.00	4.00
no_frame	0.750	0.750	0.750	10.50	4.00
no_justification	0.500	0.500	0.500	13.25	4.00
event_only	0.000	0.000	0.000	16.00	4.00

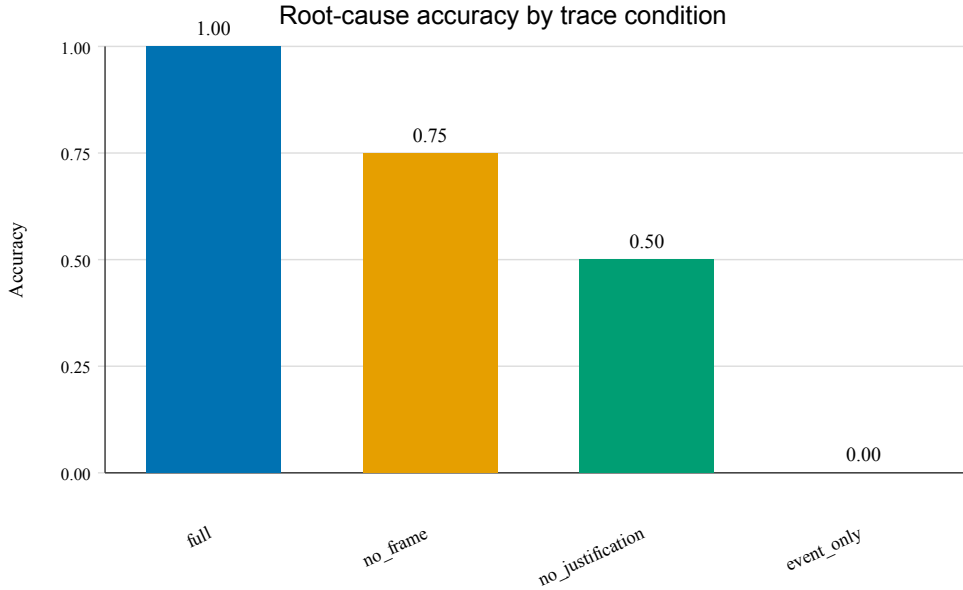


Figure 1: Generic root-cause diagnosis accuracy by trace condition.

3. **Facet conflict:** two facets conflict and the response follows one while ignoring the other.
4. **Persona drift:** the response introduces an identity inconsistent with the facet set.
5. **Response-style violation:** the response ignores the selected interaction-style facet.

The benchmark evaluates four conditions: full trace, no facet weights, no facet prompts, and event-only trace. We ran 200 cases with seed 42:

```
npx ts-node --transpile-only examples/facet-frame-faithfulness-benchmark.ts --seed=42 --cases=200
```

Table 2: Facet-frame faithfulness results. After fixing the omission-control bug, full traces outperform ablations in this deterministic sensitivity harness. Removing facet weights harms diagnosis of weighting errors. Removing facet prompts and expected response markers sharply reduces coverage.

Trace condition	Accuracy	Macro F1	Coverage	Events inspected	Trace events
full	0.865	0.898	0.865	18.60	7.80
no_facet_weights	0.665	0.678	0.735	22.80	7.80
no_facet_prompts	0.400	0.400	0.400	21.60	7.80
event_only	0.000	0.000	0.000	31.20	7.80

A representative facet-misweighting case illustrates the diagnostic path. The persona was expected to evaluate a premium AI subscription while constrained by crisis-flow financial outlook and a time-poverty facet. The trace showed **time_poverty** with observed weight 0.18 against expected weight 0.78. The response gave a long, careful evaluation despite the time constraint. The diagnostic procedure predicted **facet_misweighting** after inspecting the facet contribution event.

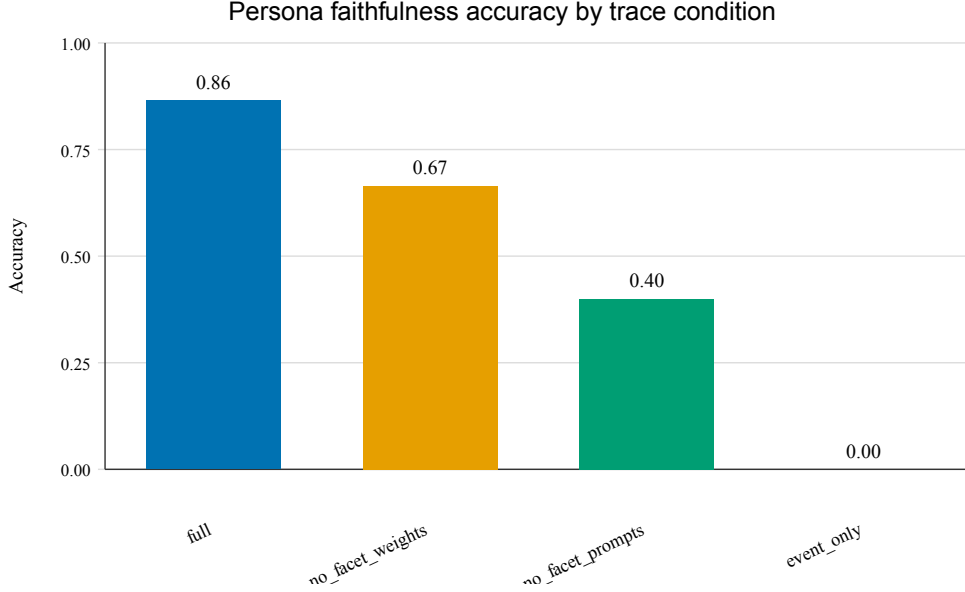


Figure 2: Facet-frame faithfulness diagnosis accuracy by trace condition.

10 Facet Faithfulness Audit Service Experiment

The benchmark above motivated a concrete implementation: **FacetFaithfulnessAuditService**. The service takes a persona response, selected facets, rendered prompt provenance, model metadata, self-reported confidence, and optional counterfactual probes. It returns a facet influence ledger, marker coverage, contradictions, counterfactual sensitivity, calibration score, final faithfulness score, primary diagnosis, causal facets, evidence, and recommended debug actions.

The service implements the audit hooks proposed by the framework: selected facet IDs and values, taxonomy versions, prompt hashes, configured and runtime weights, frame composition, suppressed facets, inferred interaction terms, expected markers, observed markers, forbidden marker contradictions, confidence calibration, and counterfactual sensitivity. A Vurvey adapter maps PM facet values, facet codes, **headerPrompt**, **detailPrompt**, confidence scores, taxonomy versions, construction-manifest weights, and rendered prompt provenance into this audit input shape.

We evaluated the audit service on eight hand-constructed Vurvey-style persona cases covering faithful responses, prompt omission, facet misweighting, facet conflict, persona drift, response-style violation, and low counterfactual sensitivity. The goal of this experiment is not to prove external predictive validity. It tests whether the implemented service can recover failure causes from increasingly rich evidence conditions.

Table 3: Facet faithfulness audit service results. Prompt-only and facet-list evidence are insufficient for reliable diagnosis. Adding weights improves diagnosis, and full AEF evidence, including prompt provenance and counterfactual probes, recovers all eight handcrafted fixture labels. These are fixture labels rather than external deployment labels.

Condition	Accuracy	Faithfulness	Marker cov.	Counterfact.	Calibration
prompt_only	0.125	0.162	0.000	0.500	0.125
facet_list	0.125	0.382	0.000	0.500	0.125
facet_weights	0.750	0.526	0.281	0.500	0.406
full_aef	1.000	0.536	0.281	0.660	0.406

The experiment exposed an important design lesson. Mean faithfulness score is not the same as diagnostic accuracy. The full audit condition intentionally surfaces more missing markers and calibration penalties than the weaker conditions, so the mean score can remain conservative even while diagnosis improves. This is desirable for an audit system: it should not inflate trust simply because it sees more evidence. The useful output is the diagnosis and debug action, not a single high score.

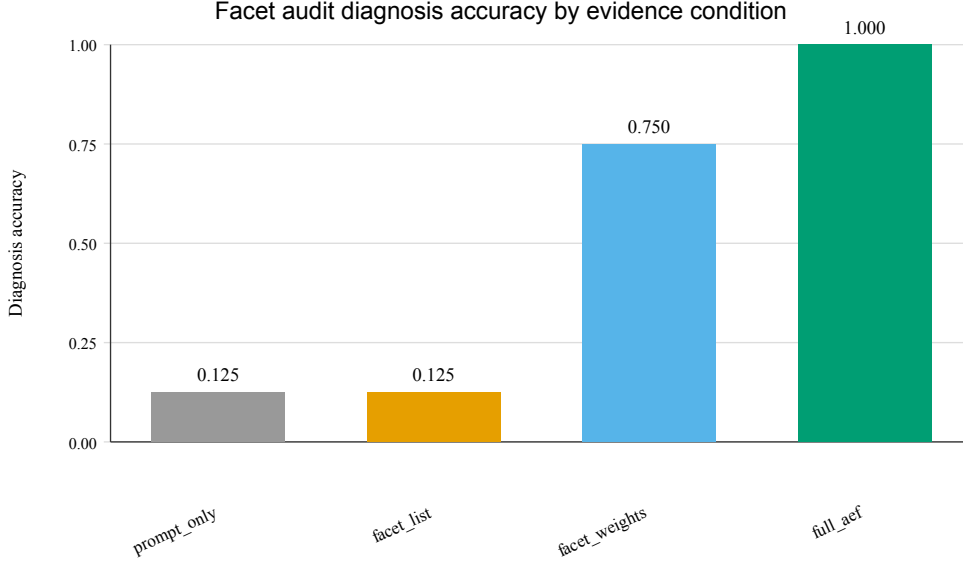


Figure 3: Facet audit diagnosis accuracy by available evidence condition.

In the faithful fixture, the ledger recorded five selected facets, prompt inclusion for each facet, prompt hashes, configured and runtime weights, and interaction terms such as `financial_outlook:crisis_flow x adoption_curve_position:innovator`. The service correctly marked the response as faithful while still listing missing secondary markers, such as explicit novelty language. In the prompt omission fixture, the rendered prompt omitted `financial_outlook`; the service diagnosed `facet_prompt_omission` and recommended verifying prompt assembly. In the persona drift fixture, the response introduced an unsupported enterprise CTO identity with unlimited budget; the service diagnosed `persona_drift` and recommended regenerating or constraining the persona biography.

11 Live Vurvey Counterfactual Generation

The fixture experiments above still use controlled responses. To test the framework against the actual Vurvey persona generation path, we added a Vurvey API counterfactual runner that calls `PersonaAnswerGenerator.generate` for a baseline persona and again for a counterfactual persona variant. The counterfactual variant is produced by applying a facet intervention to the persona’s PM facet values before generation. This experiment therefore uses the live survey-simulation prompt path, the selected persona facets, the generated system prompt, and the configured Gemini survey-simulation model rather than simulated counterfactual scores.

We first extracted read-only facet fixtures from staging and production. Staging produced a representative fixture with 25 high-facet-count personas, 15 published molds, and 157 distinct facet codes. Production produced 25 high-facet-count personas and 71 distinct facet codes; the production database currently contains one draft base mold and no mold facet configurations, so production mold distributions are not reported. Common staging facet codes include `adoption_curve`, `agreeableness`, `brand_relationship_orientation`, `caregiving_load`, and `disposable_income`. Common production facet codes include `adoption_curve`, `age`, `agreeableness`, `anxiety_level`, `challenge_mindset`, and several China-specific ecological and commercial context facets.

We then ran the live counterfactual experiment on three staging personas using the stimulus: *Evaluate a premium AI subscription that takes setup time and charges monthly*. Each case selected the first available high-impact facet from `disposable_income`, `financial_outlook`, `time_poverty`, `risk_tolerance`, `adoption_curve`, or `stress_level`, removed that facet, regenerated the response, and compared the baseline and counterfactual outputs.

The qualitative outputs were informative. One staging persona responded: “A monthly cost is already a stretch for anything extra, and if it takes time to set up? Forget about it.” After the `disposable_income` facet was removed, the generated response still referenced money and family constraints, indicating that

Table 4: Live Vurvey counterfactual generation results on three staging personas. The counterfactual sensitivity is high, indicating that removing selected facets changes generated outputs. Marker coverage remains low, showing that current marker rules are too sparse and literal for real persona responses.

Cases	Faithfulness	Counterfact. sensitivity	Marker cov.	Primary diagnosis
3	0.579	0.886	0.248	facet_selection_gap (3/3)

similar constraints are redundantly represented in other facets or in the generated biography. Another persona with high disposable income treated the product as plausible if it saved time, while the counterfactual version became more value-conditional. These outcomes show that facets do influence the live generation path, but effects are distributed across multiple identity layers.

The main finding is that real counterfactual sensitivity is measurable, but marker coverage is currently the limiting factor. The audit service correctly avoided overclaiming faithfulness: all three live cases were diagnosed as `facet_selection_gap` because the expected marker catalog did not cover enough of the real facet semantics and because removing one facet did not remove all related behavioral constraints from the biography and adjacent facets. This is a useful failure mode. It shows that the product improvement should be richer facet-level marker metadata, interaction-aware expectations, and persisted construction manifests. The implementation now records such manifests at persona initialization time and uses them at survey-simulation runtime, closing the main traceability gap identified by the live counterfactual experiment.

12 Discussion

The generic benchmark shows that structured epistemic traces can expose simple injected failures when the trace contains the relevant fields. The facet benchmark is more realistic and harder. Full traces recover many failures, but not all. This is useful: persona faithfulness cannot be reduced to checking that a prompt contains the right words. It requires tracing facet selection, prompt rendering, frame composition, expected behavioral markers, response content, and sometimes counterfactual or latent evidence.

For Vurvey-style synthetic personas, AEF suggests several concrete audit hooks:

1. Record the selected facet IDs, value IDs, codes, taxonomy versions, and prompt hashes used for a response.
2. Record facet weights and interaction terms used to compose the active frame.
3. Record expected response markers derived from selected facets, such as cost objection, peer evidence request, time constraint, or response-style marker.
4. Run counterfactual probes that swap one high-impact facet while holding the rest constant.
5. Track self-rated confidence and compare it with evaluator judgments and response consistency.
6. When model internals are available, add representation-level probes for persona-relevant directions or sparse features.

13 Limitations

The benchmarks use controlled synthetic failures with known labels and deterministic heuristic evaluators. They establish that the implementation can carry and expose useful diagnostic fields, not that all synthetic personas are valid or predictive in deployment, and not that the reported synthetic accuracies estimate real-world performance. The facet benchmark should be extended with noisy traces, hidden labels, independent evaluators, human debugging studies, live product prompt paths, and held-out response data.

The framework also depends on the quality of facet definitions, prompt fragments, weights, and constraints. If the facet graph itself encodes weak theory or biased assumptions, AEF will make those assumptions more visible but will not make them correct. Detailed observer logs can expose sensitive

persona or respondent data, so production use requires careful retention, access control, and redaction policy.

14 Code and Data Availability

The reference implementation is available at <https://github.com/jroell/agentive-epistemology-framework> under the MIT license. The implementation includes `FacetFaithfulnessAuditService`, the generic root-cause benchmark, the facet-frame benchmark, the facet audit service experiment, public benchmark documentation, and a Vurvey construction-manifest/runtime-audit integration. Generated outputs are written under the repository’s results directory. A Vurvey API branch named `AEF-implementation` contains the product-facing adapter that maps PM facet values and construction-manifest weights into the audit input shape, attaches AEF audit summaries during survey simulation, and includes the live Vurvey counterfactual generation runner. A Vurvey Python branch named `AEF-implementation` persists AEF construction manifests into `ai_personas.facetConfigs` during mold-based persona generation.

15 Conclusion

The Agentive Epistemology Framework provides an audit architecture for autonomous agents and synthetic personas. By making beliefs, confidence, justifications, frames, facets, and observer events first-class constructs, AEF supports debugging not only what an agent said, but why the system should have expected that behavior from its configured epistemic state. The facet extension connects this audit layer to synthetic human simulation: facets become inspectable contributors to frame composition, and persona faithfulness becomes a measurable, diagnosable property rather than a matter of surface plausibility.

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